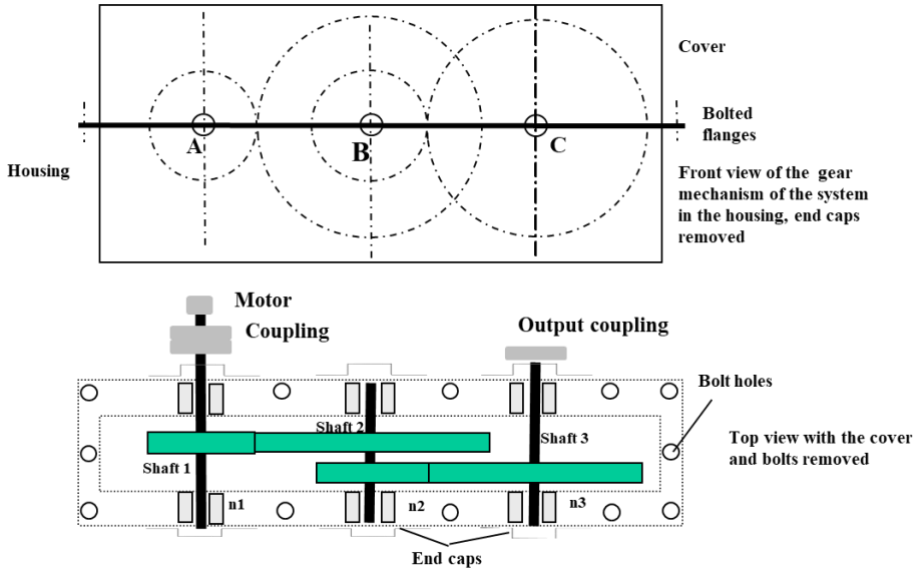
**ME 315**
**THEORY OF MACHINES – DESIGN OF ELEMENTS**
Spring 2026

**Design Project**
Group submission



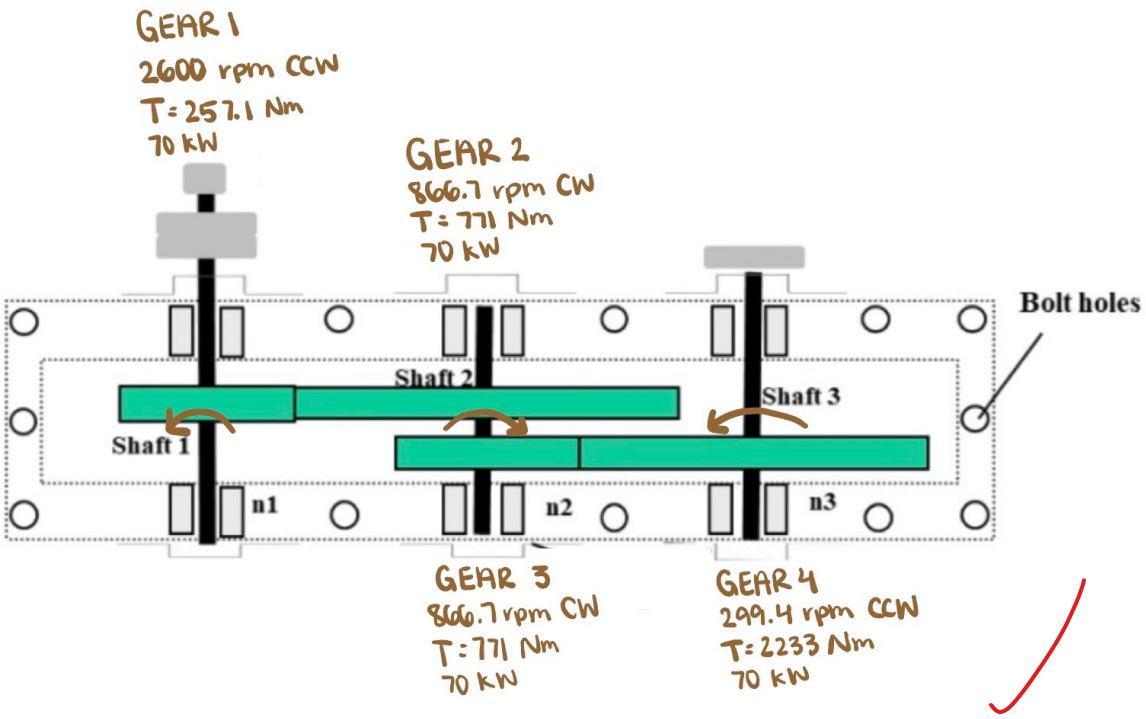

**Group members**

| | | |
|---|---|---|
| Jordan Chu | Shaft system 1 | Total 86 /95 |
| Crystal Cooper | Shaft system 2 | Total 86 /95 |
| Sujee Rubio | Shaft system 3 | Total 86 /95 |

**Grading scale**

| | |
|---|---|
| On-time step dues | (5) |
| Structural design, drawings | 35 /40 (assembly, two parts, dimensions, tolerances, etc.) |
| Design analysis | 37 /40 (shaft, gear(s), bearings) |
| Report | 14 /15 (view quality, cover sheet, page numbers, clarity, logic, conciseness) |

# Basic Gear Box Diagram



| Item | Gear 1 | Gear 2 | Gear 3 | Gear 4 |
|---|---|---|---|---|
| Shaft | A | B | B | C |
| Teeth number, $N$ | 17 | 51 | 19 | 55 |
| Speed, $n$ (rpm) | 2600 | 866.7 | 866.7 | 299.4 |
| Power, $P$ (kW) | 70 | 70 | 70 | 70 |
| Torque, $T$ (N·m) | 257.1 | -771 | 771 | -2233 |
| Transmission error | 0.20% | .020% | .020% | 0.20% |
| Module, $m$ (mm) | 5 | 5 | 5 | 5 |
| Pressure angle, $\phi$ (°) | 20 | 20 | 20 | 20 |
| Pitch diameter, $d_p$ (mm) | 85 | 255 | 95 | 275 |
| Face width, $b_w$ (mm) | 50 | 50 | 75 | 75 |
| Tangential force, $W_t$ (N) | 6047.1 | 6047.1 | 16231.6 | 16231.6 |
| Radial force, $W_r$ (N) | 2201 | -2201 | 5907.8 | -5907.8 |
| Material | AISI 1045 | AISI 1045 | AISI 1045 | AISI 1045 |
| Hardness | 450 HB | 200 HB | 450 HB | 200 HB |
| FOS $n_{s,b}$; $n_{s,c}$ | 2.72; 2.14 | 1.54; 2.47 | 1.36; 2.31 | 1.37; 3.40 |
| Centerline distance (mm) | $C_{AB} = 170$ | | $C_{BC} = 185$ | |
| Total A–C distance (mm) | 355 | | | |

Table 1: Gear train and gear parameters.

| Item | Shaft A | Shaft B | Shaft C |
|---|---|---|---|
| Material | AISI 1045 CD | AISI 1045 CD | AISI 1045 CD |
| Speed, $n$ (rpm) | 2600 | 866.7 | 299.4 |
| Maximum torque, $T$ (N·m) | 257.1 | 771 | 2233 |
| Maximum bending moment, $M$ (N·m) | 232.8 | 818.8 | 699.3 |
| Critical cross section | Coupling keyway | Gear 3 keyway | Gear keyway |
| Goodman FoS, $n_f$ | 1.91 | 2.68 | 2.29 |
| Yield FoS, $n_y$ | 2.29 | 5.29 | 3.07 |
| Expected life | Infinite | Infinite | Infinite |

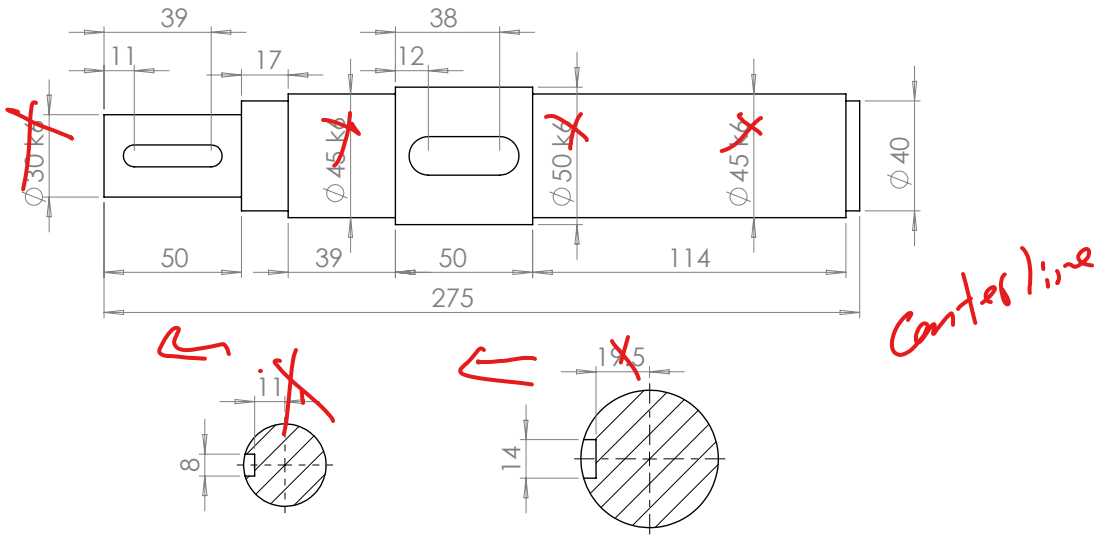
Table 2: Shaft design parameters.

| Item | Shaft A Bearing 1 | Shaft A Bearing 2 | Shaft B Bearing 1 | Shaft B Bearing 2 | Shaft C Bearing 1 | Shaft C Bearing 2 |
|---|---|---|---|---|---|---|
| Designation | SKF 6409 | SKF 6209 | SKF 6316 | SKF 6313 | SKF 6313 | SKF 6313 |
| Radial load, $F_r$ (N) | 4704 | 1731 | 413.1 | 3293.7 | 5420.9 | 11852.4 |
| Thrust load, $F_a$ (N) | 0 | 0 | 0 | 0 | 0 | 0 |
| Bore ID (mm) | 45 | 45 | 65 | 65 | 65 | 65 |
| OD (mm) | 120 | 85 | 140 | 140 | 140 | 140 |
| Width (mm) | 29 | 19 | 33 | 33 | 33 | 33 |
| $C$ (kN) | 76.1 | 33.2 | 97.5 | 97.5 | 97.5 | 97.5 |
| $C_0$ (kN) | 45.0 | 21.6 | 60.0 | 60.0 | 60.0 | 60.0 |
| Bearing life (hr) | 27,100 | 45,200 | 8511.5 | 8511.5 | 323,700 | 30,988 |
| Bearing change needed? | No | No | Yes (once) | Yes (once) | No | No |

Table 3: Bearing selection and life for all three shafts. Shaft B bearing selection pending.

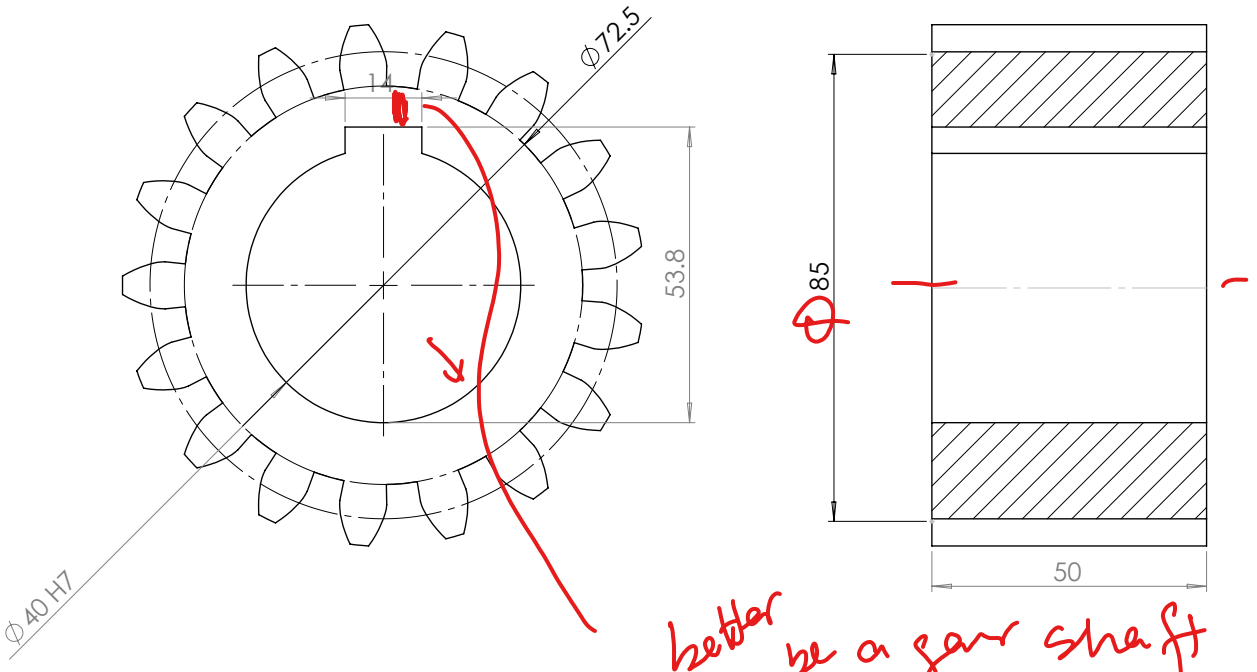
| Parameter | Value |
|---|---|
| Power, $P$ (kW) | 70 |
| Input speed, $n_{in}$ (rpm) | 2600 |
| Output speed, $n_{out}$ (rpm) | 299.4 |
| Overall speed ratio, $i_{tot}$ | 8.684 |
| Overall design FoS (min.) | $n_f = 1.36$ (Gear 3 bending, Shaft B) |
| Bearing changes required? | Yes — Shaft B bearings replaced once at ∼year 3 (7280 hr) |
| Oil change interval | Every 5 years (14.560 hr) or at bearing replacement, whichever comes first |
| Additional maintenance notes | All shafts have infinite fatigue life. Minimum Goodman FoS across all shafts is 1.91 (Shaft A coupling keyway). Minimum gear bending FoS is 1.36 (Gear 3). Inspect coupling and keyway fits at each bearing service interval. |

Table 4: Overall transmission design results and maintenance schedule.




| | | | | | | | |
|---|---|---|---|---|---|---|---|
| UNLESS OTHERWISE SPECIFIED:<br>DIMENSIONS ARE IN MILLIMETERS<br>SURFACE FINISH:<br>TOLERANCES:<br>LINEAR:<br>ANGULAR: | | FINISH: | | DEBURR AND BREAK SHARP EDGES | | DO NOT SCALE DRAWING | REVISION |
| | NAME | SIGNATURE | DATE | | | TITLE: | |
| DRAWN | Jordan Chu | | | | | Shaft A | |
| CHK'D | | | | | | | |
| APPV'D | | | | | | | |
| MFG | | | | | | | |
| Q.A | | | | MATERIAL: | | DWG NO. | A4 |
| | | | | | | Shaft A | |
| | | | | WEIGHT: | | SCALE:1:5 | SHEET 1 OF 1 |

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| | |
|---|---|
| Module, m | 5 mm |
| Pressure Angle | 20 deg. |
| Number of Teeth | 17 |
| Pitch Diameter | 85 mm |
| Addendum, a | 5 mm |
| Dedendum, b | 6.25 mm |
| Clearance, c | 1.25 mm |
| Material | AISI 1045 |
| Heat Treatment | None |

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING | REVISION

| | NAME | SIGNATURE | DATE |
|---|---|---|---|
| DRAWN | Jordan Chu | | |
| CHK'D | | | |
| APPV'D | | | |
| MFG | | | |
| Q.A | | | |

MATERIAL:

WEIGHT:

TITLE:

# Shaft A - Gear 1

DWG NO.

Gear 1

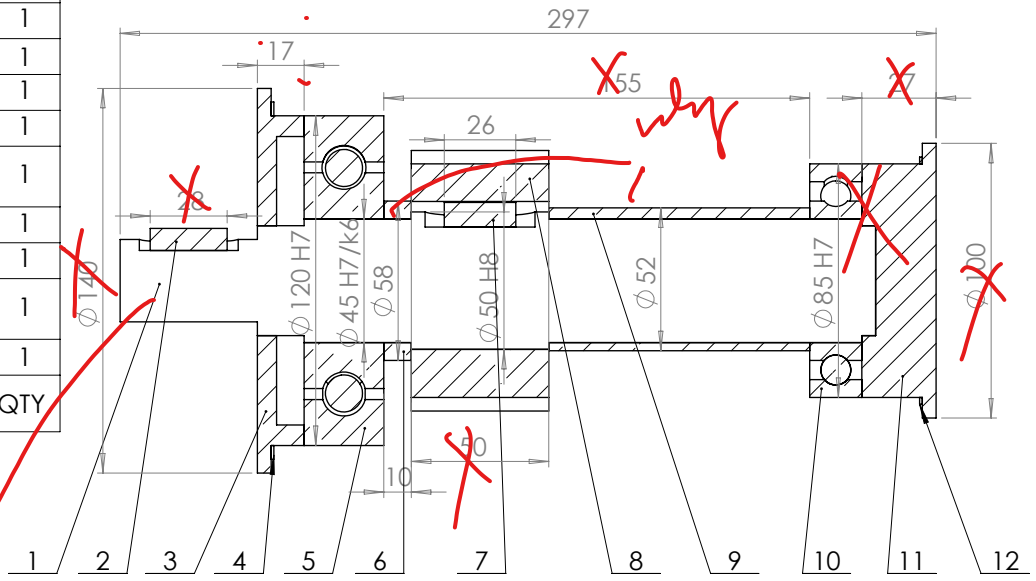
A4

SCALE:1:2

SHEET 1 OF 1

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| ITEM NO. | PART NUMBER | Material | QTY |
|---|---|---|---|
| 12 | Blind Gasket | Rubber | 1 |
| 11 | Blind Endcap | AISI 1018 | 1 |
| 10 | SKF 6209 | Bearing Steel | 1 |
| 9 | Spacer, 125mm | AISI 1018 | 1 |
| 8 | Gear | AISI 1045 | 1 |
| 7 | Gear Key | AISI 1045 | 1 |
| 6 | Spacer, 5mm | AISI 1018 | 1 |
| 5 | SKF 6409 | Bearing Steel | 1 |
| 4 | Input Gasket | Rubber | 1 |
| 3 | Input Endcap | AISI 1018 | 1 |
| 2 | Coupling Key | AISI 1045 | 1 |
| 1 | Shaft | AISI 1045 CD | 1 |




UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING | REVISION

| | NAME | SIGNATURE | DATE |
|---|---|---|---|
| DRAWN | Jordan Chu | | |
| CHK'D | | | |
| APPV'D | | | |
| MFG | | | |
| Q.A | | | |

TITLE:

# Shaft A Assembly

MATERIAL:

DWG NO. Shaft_A_ASM

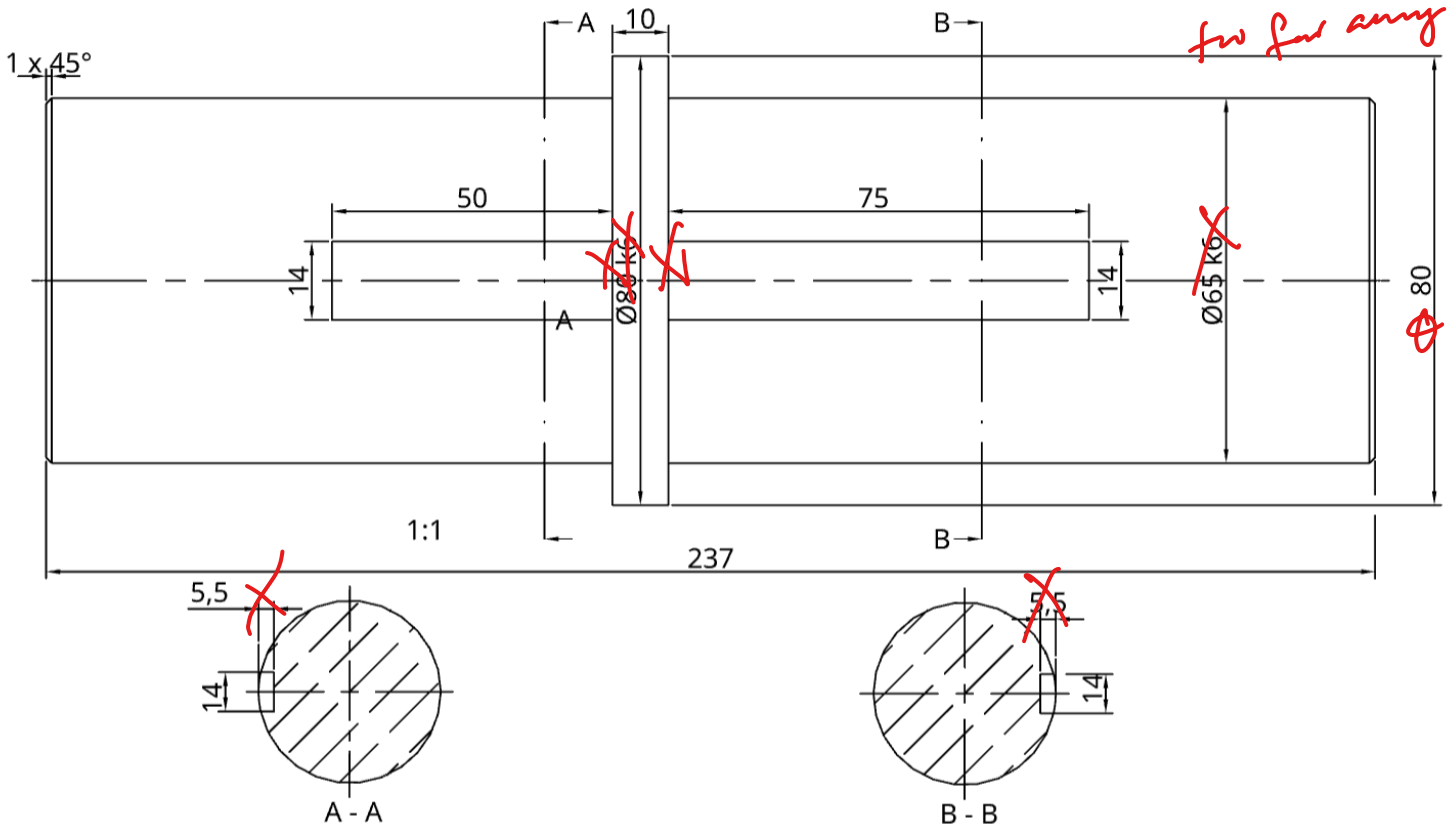
A4

WEIGHT:

SCALE:1:5

SHEET 1 OF 1

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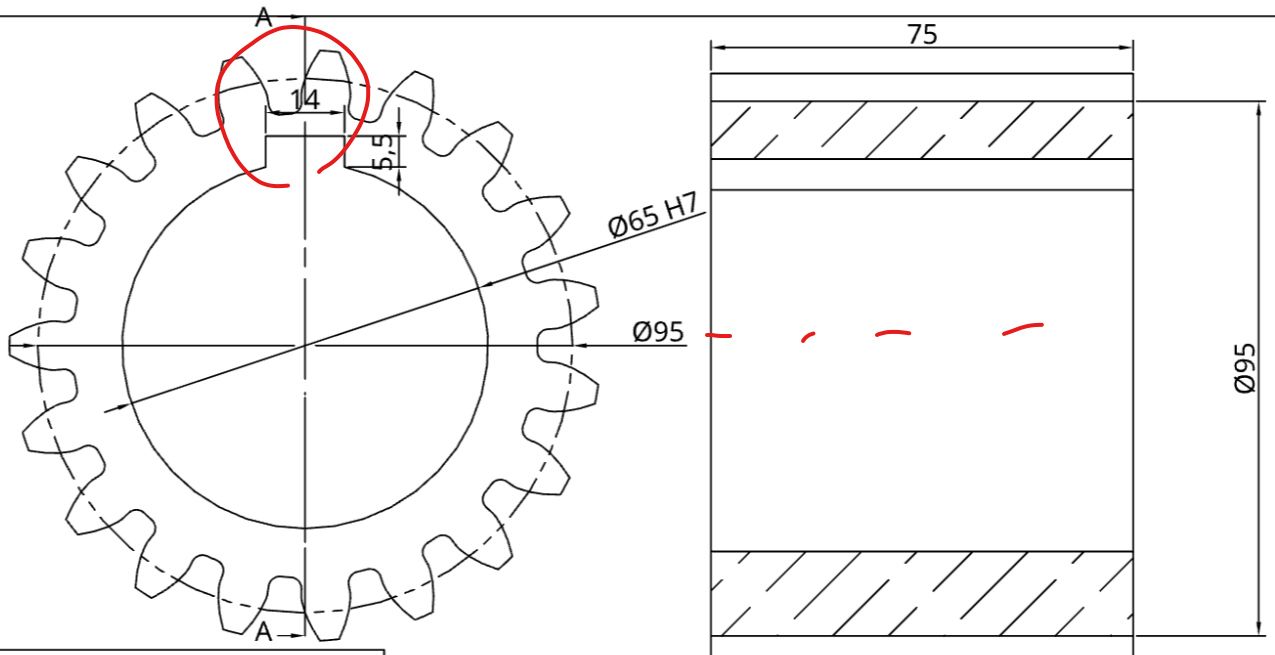



| | | | | |
|---|---|---|---|---|
| UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS | | NAME | SIGNATURE | DATE |
| ANGULAR = ± ° | DRAWN | CRYSTAL COOPER | | 2026-06-09 |
| SURFACE FINISH √ | CHECKED | | | |
| | APPROVED | — | | — |
| DO NOT SCALE DRAWING | | | | |
| BREAK ALL SHARP EDGES AND REMOVE BURRS | | | | |
| FIRST ANGLE PROJECTION | MATERIAL | | FINISH | |

TITLE: Shaft B

| SIZE | DWG NO. | REV. |
|---|---|---|
| A4 | Shaft B | - |

| SCALE | WEIGHT | SHEET |
|---|---|---|
| 1:1 | | 1 of 3 |




| | |
|---|---|
| Module | 5 mm |
| Pressure angle | 20 degrees |
| Number of Teeth | 19 |
| Addendum | 5 mm |
| Dedendum | 6.25 mm |
| Pitch Circle Diameter | 95 mm |
| Clearance | 1.25 mm |
| Material | AISI 1045 CD |
| Heat Treatment | Induction Hardened, ~450 HB |

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS
ANGULAR = ± °
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
FIRST ANGLE PROJECTION

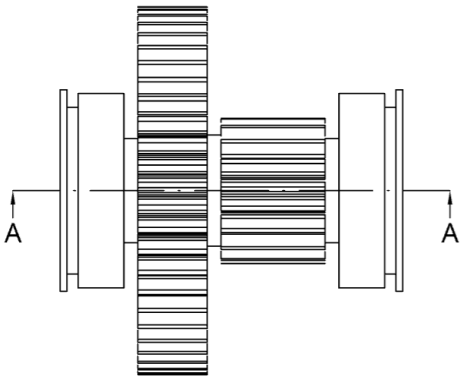
| | NAME | SIGNATURE | DATE |
|---|---|---|---|
| DRAWN | CRYSTAL COOPER | | 2026-04-29 |
| CHECKED | | | |
| APPROVED | --- | | --- |

MATERIAL
FINISH

TITLE
Gear 3 Drawing

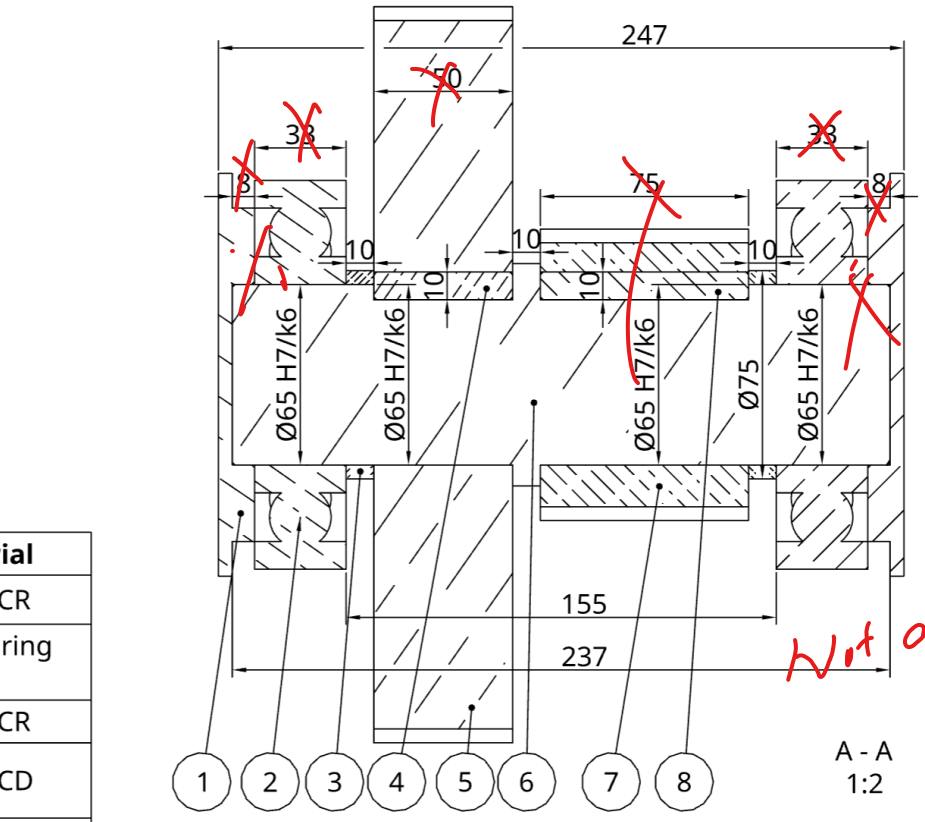
SI A4 | DWG NO. Gear 3 Drawing | REV. -

SCALE 1:1 | WEIGHT | SHEET 2 of 3




1:4

| Item | Qty | Part Name | Material |
|---|---|---|---|
| 1 | 2 | End Cap | AISI 1018 CR |
| 2 | 2 | SKF 6313 Bearing | 52100 Bearing Steel |
| 3 | 2 | Spacer | AISI 1018 CR |
| 4 | 1 | 50 mm Gear Key | AISI 1045 CD |
| 5 | 1 | Gear 2 | AISI 1045 CD |
| 6 | 1 | Shaft B | AISI 1045 CD |
| 7 | 1 | Gear 3 | AISI 1045 CD |
| 8 | 1 | 75 mm Gear Key | AISI 1045 CD |




A - A
1:2

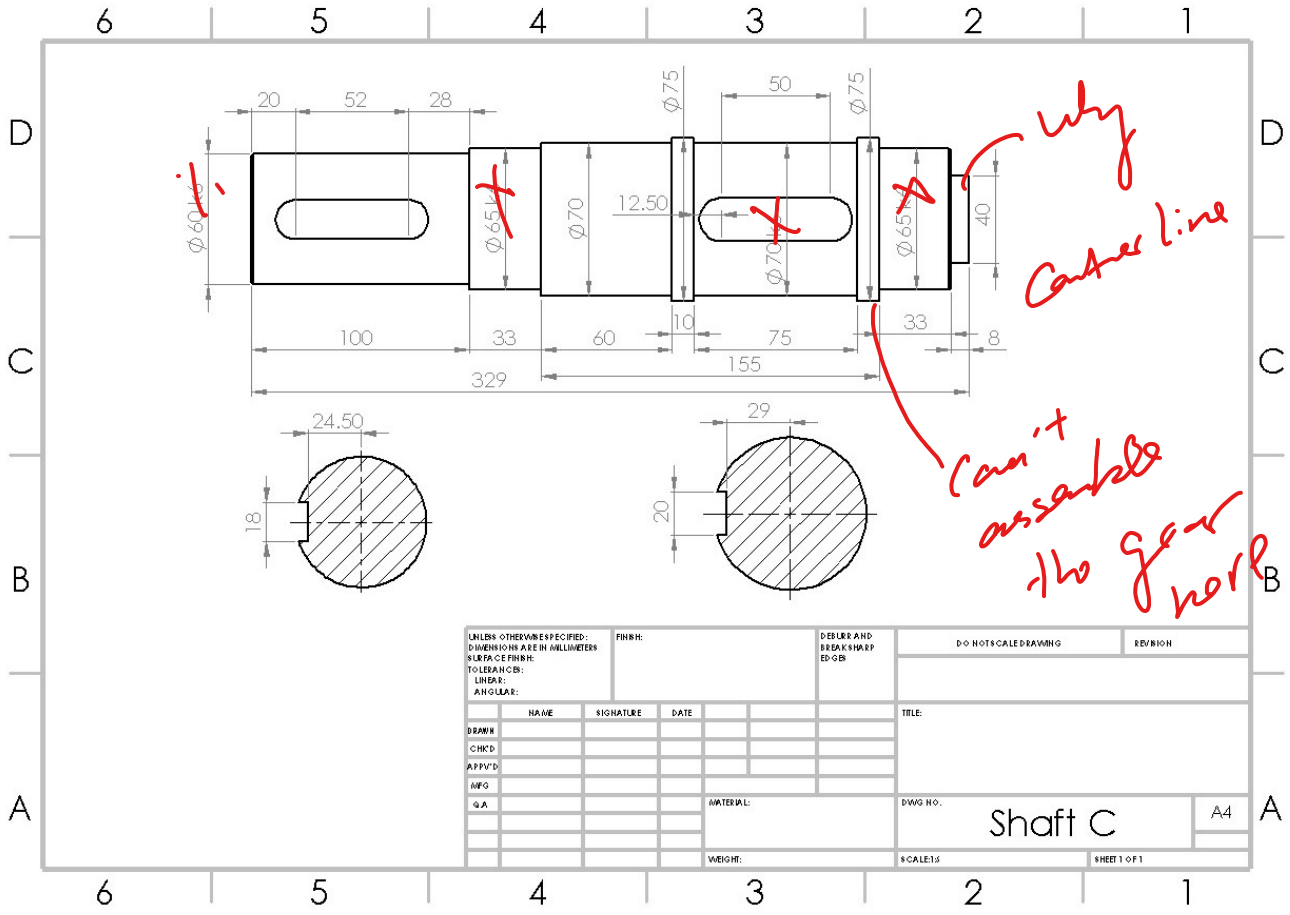
Not all KG

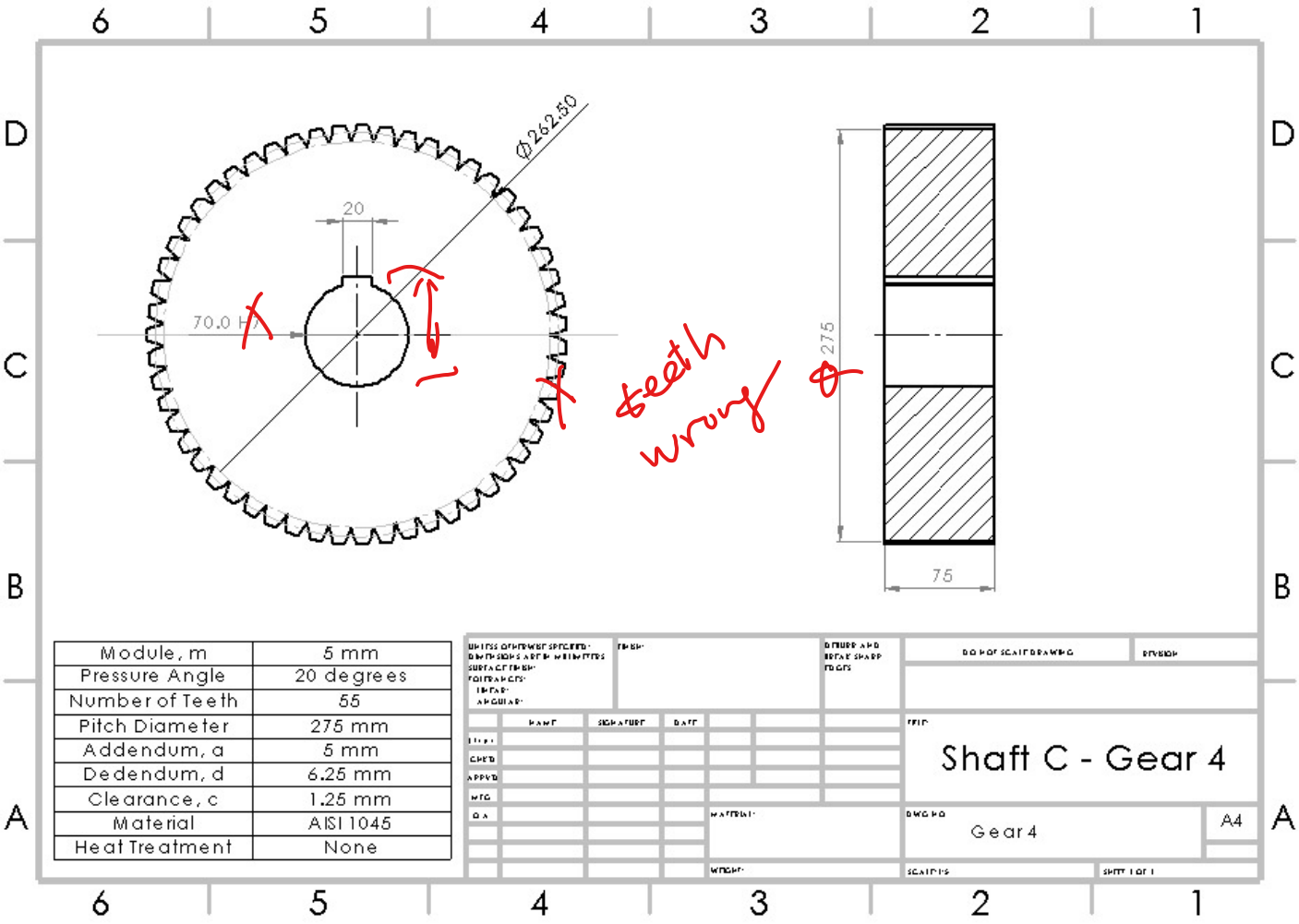
| UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS | | NAME | SIGNATURE | DATE |
|---|---|---|---|---|
| ANGULAR = ± ° | DRAWN | CRYSTAL COOPER | | 2026-06-09 |
| SURFACE FINISH | CHECKED | | | |
| | APPROVED | — | | — |
| DO NOT SCALE DRAWING | | | | |
| BREAK ALL SHARP EDGES AND REMOVE BURRS | | | | |
| FIRST ANGLE PROJECTION | MATERIAL | | FINISH | |

TITLE: Shaft B Assembly

| SI | DWG NO. | REV. |
|---|---|---|
| A4 | Shaft B Assembly | - |

| SCALE | WEIGHT | SHEET |
|---|---|---|
| | | 3 of 3 |


20
52
28
Ø75
50
Ø75
Ø60
Ø65
Ø70
12.50
Ø70
Ø65
40
100
33
60
10
75
33
8
155
329
24.50
18
29
20
why
Center line
can't assemble the gear
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:
FINISH:
DEBURR AND BREAK SHARP EDGES
DO NOT SCALE DRAWING
REVISION
NAME
SIGNATURE
DATE
DRAWN
CHK'D
APPV'D
MFG
Q.A
TITLE:
MATERIAL:
DWG NO.
Shaft C
A4
WEIGHT:
SCALE:1:5
SHEET 1 OF 1



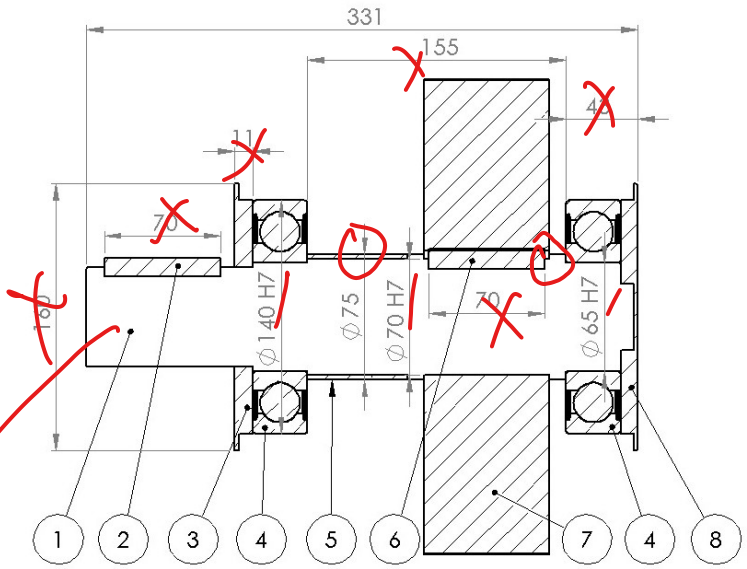

| | |
|---|---|
| Module, m | 5 mm |
| Pressure Angle | 20 degrees |
| Number of Teeth | 55 |
| Pitch Diameter | 275 mm |
| Addendum, a | 5 mm |
| Dedendum, d | 6.25 mm |
| Clearance, c | 1.25 mm |
| Material | AISI 1045 |
| Heat Treatment | None |

Shaft C - Gear 4

Gear 4

A4

| ITEM NO. | PART NUMBER | Material | QTY. |
|---|---|---|---|
| 1 | Shaft C | AISI 1045 CD | 1 |
| 2 | Coupling Key | AISI 1045 | 1 |
| 3 | Output Endcap | AISI 1018 | 1 |
| 4 | SKF 6313 | Bearing Steel | 2 |
| 5 | Spacer, 60mm | AISI 1018 | 1 |
| 6 | Gear Key | AISI 1045 | 1 |
| 7 | Gear 4 | AISI 1045 | 1 |
| 8 | Blind Endcap | AISI 1018 | 1 |




UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

| | NAME | SIGNATURE | DATE |
|---|---|---|---|
| DRAWN | | | |
| CHK'D | | | |
| APP'D | | | |
| MFG | | | |
| Q.A | | | |

TITLE:

# Shaft C Assembly

MATERIAL:

DWG NO.

A4

WEIGHT:

SCALE:1:10

SHEET 1 OF 1

# Appendix A — Shaft System A

*Jordan Chu*

## A.1 Shaft Analysis

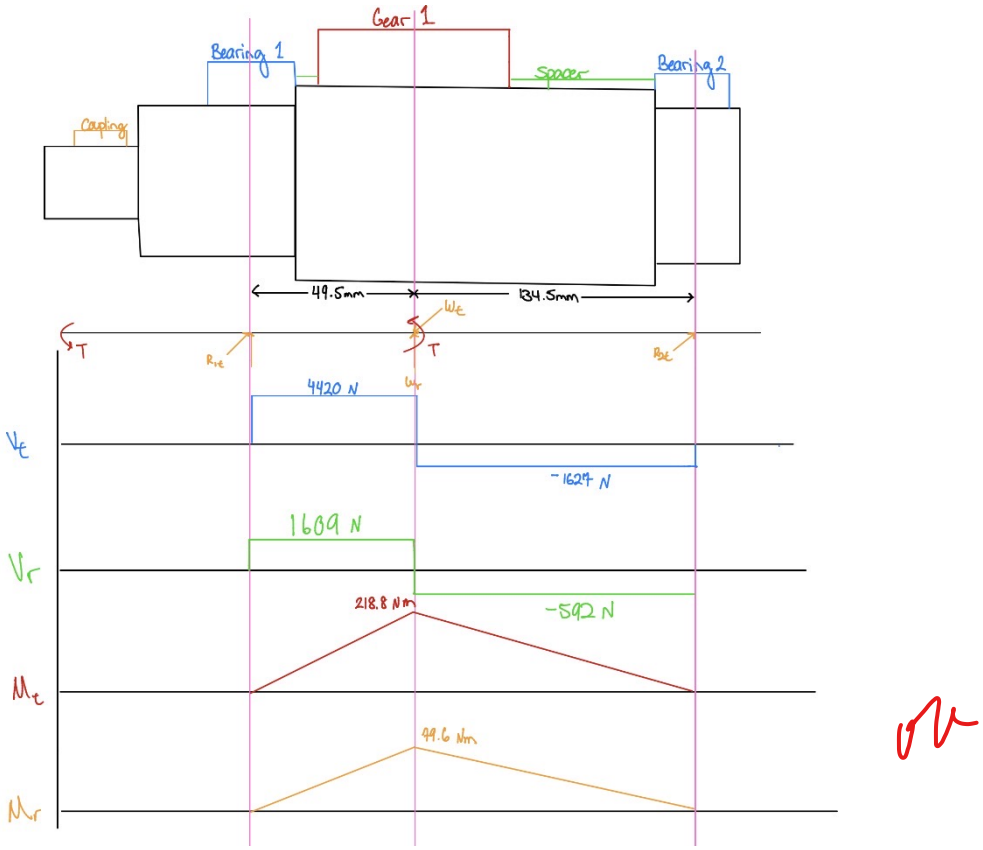



Figure 1: Shaft A FBD, shear force, and bending moment diagrams.

Shaft A is the input shaft, driven by the motor through a coupling at 2600 rpm. It carries Gear 1, the 17-tooth pinion of the first gear pair. The gear center sits 49.5 mm from the input-side bearing center (center-to-center span 184 mm; bearing inner faces 155 mm apart to match the gearbox housing). Material is AISI 1045 cold-drawn steel. Because the drive starts and stops daily, the transmitted torque fluctuates between zero and its maximum, so it is treated as having equal mean and alternating components.

**Given:**

$$N_1 = 17 \qquad n_1 = 2600 \text{ rpm} \qquad H = 70 \text{ kW} \qquad m = 5 \text{ mm} \qquad \phi = 20^\circ \tag{1}$$

$$S_{ut} = 550 \text{ MPa} \qquad S_y = 310 \text{ MPa} \qquad n_s \geq 1.3 \tag{2}$$

**Torque:**

$$T = \frac{60H}{2\pi n_1} = \frac{60(70000)}{2\pi(2600)} = 257.1 \text{ N} \cdot \text{m} \tag{3}$$

**Pitch Diameter and Gear Forces:**

$$d_1 = N_1 m = 17(5) = 85 \text{ mm} \tag{4}$$

$$W_t = \frac{2T}{d_1} = \frac{2(257.1)}{0.085} = 6047 \text{ N} \qquad W_r = W_t \tan\phi = 2201 \text{ N} \qquad W_a = 0 \tag{5}$$

**Step 1 — Minimum Shaft Diameter (torque-based):**

$$\tau_{DET} = \frac{0.577\, S_y}{n_s} = \frac{0.577(310)}{1.3} = 137.6 \text{ MPa} \tag{6}$$

$$d_{min} \geq \sqrt[3]{\frac{16T}{\pi\, \tau_{DET}}} = \sqrt[3]{\frac{16(257.1)}{\pi(137.6 \times 10^6)}} = 21.2 \text{ mm} \tag{7}$$

Keyway correction (+5%): $d = 21.2(1.05) = 22.3\,\text{mm}$.

**Endurance Limit (Marin factors, ground surface):**

$$S'_e = 0.5\, S_{ut} = 275 \text{ MPa} \qquad k_f = 1.58\, S_{ut}^{-0.085} = 0.917 \tag{8}$$

$$k_s = 1.189\, d^{-0.112} \;\Rightarrow\; k_s = 0.767 \;(d = 50), \quad k_s = 0.812 \;(d = 30) \tag{9}$$

$$k_r = 0.82 \;(99\%) \qquad k_t = 1 \qquad k_m = 1 \tag{10}$$

$$S_e = k_f k_s k_r k_t k_m S'_e = 159.9 \text{ MPa } (\varnothing 50), \quad 169.3 \text{ MPa } (\varnothing 30) \tag{11}$$

**Stress Concentration (keyway):**

$$K_f = K_{fs} = 1.61 \tag{12}$$

From Spotts & Shoup (1998) for medium-carbon steel (SAE 1045, $S_{ut} \approx 550\,\text{MPa}$), per the project guidelines.

**Bearing Reaction Forces:** With the gear off-center ($a = 49.5\,\text{mm}$, $b = 134.5\,\text{mm}$, $L = 184\,\text{mm}$):

$$R_{1t} = W_t \frac{b}{L} = 4420 \text{ N} \qquad R_{2t} = W_t \frac{a}{L} = 1627 \text{ N} \tag{13}$$

$$R_{1r} = W_r \frac{b}{L} = 1609 \text{ N} \qquad R_{2r} = W_r \frac{a}{L} = 592 \text{ N} \tag{14}$$

$$R_1 = \sqrt{R_{1t}^2 + R_{1r}^2} = 4704 \text{ N (near)} \qquad R_2 = \sqrt{R_{2t}^2 + R_{2r}^2} = 1731 \text{ N (far)} \tag{15}$$

**Bending Moment at the Gear:**

$$M_t = R_{1t}\, a = 218.8 \text{ N} \cdot \text{m} \qquad M_r = R_{1r}\, a = 79.6 \text{ N} \cdot \text{m} \tag{16}$$

$$M = \sqrt{M_t^2 + M_r^2} = 232.8 \text{ N} \cdot \text{m} \tag{17}$$

**Two critical cross sections are checked: the gear keyway and the coupling keyway. Because the torque fluctuates $0 \to T$, the torsional stress is split into equal mean and alternating parts, $\tau_m = \tau_a = \frac{1}{2}(16T/\pi d^3)$.**

**Critical Section 1 — Gear Keyway (⌀50):**

$$\sigma_a = \frac{32M}{\pi d^3} = 19.0 \text{ MPa}, \qquad \sigma_m = 0 \tag{18}$$

$$\tau_m = \tau_a = \frac{1}{2} \cdot \frac{16T}{\pi d^3} = 5.2 \text{ MPa} \tag{19}$$

$$\sigma_{VM,a} = \sqrt{(K_f \sigma_a)^2 + 3(K_{fs} \tau_a)^2} = 33.9 \text{ MPa} \tag{20}$$

$$\sigma_{VM,m} = \sqrt{\sigma_m^2 + 3(K_{fs} \tau_m)^2} = 14.6 \text{ MPa} \tag{21}$$

$$\frac{1}{n_f} = \frac{\sigma_{VM,a}}{S_e} + \frac{\sigma_{VM,m}}{S_{ut}} = \frac{33.9}{159.9} + \frac{14.6}{550} \Rightarrow n_f = 4.20 \tag{22}$$

Yield check (von Mises at peak load):

$$\sigma_{VM,\max} = \sqrt{(K_f \sigma_{\max})^2 + 3(K_{fs} \tau_{\max})^2} = \sqrt{(1.61 \cdot 19.0)^2 + 3(1.61 \cdot 10.5)^2} = 42.3 \text{ MPa} \tag{23}$$

$$n_y = \frac{S_y}{\sigma_{VM,\max}} = \frac{310}{42.3} = 7.34 \tag{24}$$

**Critical Section 2 — Coupling Keyway (⌀30):**

$$\sigma_a = 0, \qquad \tau_m = \tau_a = \frac{1}{2} \cdot \frac{16T}{\pi d^3} = 24.2 \text{ MPa} \tag{25}$$

$$\sigma_{VM,a} = \sqrt{3}\, K_{fs} \tau_a = 67.6 \text{ MPa} \qquad \sigma_{VM,m} = \sqrt{3}\, K_{fs} \tau_m = 67.6 \text{ MPa} \tag{26}$$

$$\frac{1}{n_f} = \frac{67.6}{169.3} + \frac{67.6}{550} \Rightarrow n_f = 1.91 \tag{27}$$

Yield check (von Mises at peak load):

$$\sigma_{VM,\max} = \sqrt{3}\, K_{fs} \tau_{\max} = \sqrt{3}(1.61)(48.5) = 135.2 \text{ MPa} \tag{28}$$

$$n_y = \frac{S_y}{\sigma_{VM,\max}} = \frac{310}{135.2} = 2.29 \tag{29}$$

**Governing Result:**

| | Goodman $n_f$ | Yield $n_y$ |
|---|---|---|
| Gear keyway (⌀50) | 4.20 | 7.34 |
| Coupling keyway (⌀30) | 1.91 | 2.29 |

(30)

The coupling keyway governs on both criteria. Although the gear section carries the maximum bending moment, the coupling's smaller diameter (⌀30) produces a higher torsional stress (stress $\propto 1/d^3$), making it the most critical section. The governing factors of safety ($n_f = 1.91$, $n_y = 2.29$) exceed the minimum $n_s = 1.3$, so the shaft has infinite life.

## A.2 Gear Analysis (Gear 1, AGMA)

Gear 1 is the 17-tooth spur pinion of the first pair (AISI 1045, teeth induction-hardened to $\sim$450 HB; core remains 1045). Module 5, pressure angle 20°, face width 50 mm. Analyzed by the AGMA method for both bending and contact (pitting) with 99% reliability.

**Life and Stress-Cycle Factors:**

$$N = \frac{14560(60)(2600)}{10^6} = 2271 \text{ million cycles} \tag{31}$$

$$Y_N = 1.6831\,N^{-0.0323} = 1.311 \qquad Z_N = 2.466\,N^{-0.056} = 1.600 \tag{32}$$

**Pitch Line Velocity and Dynamic Factor ($Q_v = 6$):**

$$V = \frac{\pi d_1 n_1}{60} = 11.6 \text{ m/s} \tag{33}$$

$$B = 0.25(12 - Q_v)^{2/3} = 0.825 \qquad A = 50 + 56(1 - B) = 59.8 \tag{34}$$

$$K_v = \left(\frac{A + \sqrt{200V}}{A}\right)^B = 1.628 \tag{35}$$

**Load-Distribution Factor:**

$$C_{pf} = \frac{b_w}{10\,d_1} - 0.0375 + 0.000492\,b_w = 0.046 \tag{36}$$

$$C_{ma} = 0.127 + 6.22 \times 10^{-4} b_w - 1.69 \times 10^{-7} b_w^2 = 0.158 \tag{37}$$

$$K_m = 1 + (C_{pf} + C_{ma}) = 1.204 \tag{38}$$

Other factors: $K_a = 1$ (uniform load), $K_s = K_i = K_b = 1$.

**AGMA Bending:**

$$S_b = 0.533\,HB + 88.3 = 0.533(450) + 88.3 = 328 \text{ MPa} \tag{39}$$

$$\sigma_{b,\text{allow}} = \frac{S_b\,Y_N}{K_T K_R} = \frac{328(1.311)}{(1)(1)} = 430 \text{ MPa} \tag{40}$$

$$\sigma_b = \frac{W_t}{b_w\,m\,Y_J} K_a K_s K_m K_v K_i K_b = \frac{6047}{(0.050)(0.005)(0.30)}(1.204)(1.628) = 158 \text{ MPa} \tag{41}$$

$$n_b = \frac{\sigma_{b,\text{allow}}}{\sigma_b} = \frac{430}{158} = 2.72 \tag{42}$$

**AGMA Contact:** Elastic coefficient (steel–steel, $E = 200$ GPa, $\nu = 0.285$):

$$K_E = \sqrt{E'} = \sqrt{\frac{2}{\dfrac{1-\nu_p^2}{E_p} + \dfrac{1-\nu_g^2}{E_g}}} = 4.666 \times 10^5 \ \sqrt{\text{Pa}} \tag{43}$$

Geometry factor (external mesh, $d_g = 255\,\text{mm}$):

$$I = \frac{\pi \cos\phi \, \sin\phi}{1 + \dfrac{d_p}{d_g}} = \frac{\pi \cos 20^\circ \sin 20^\circ}{1 + \dfrac{85}{255}} = 0.757 \tag{44}$$

Contact stress:

$$\sigma_c = K_E \sqrt{\frac{W_t}{b_w \, d_p} \cdot \frac{1}{I} \, K_a K_s K_m K_v} = 4.666 \times 10^5 \sqrt{\frac{6047}{(0.050)(0.085)} \cdot \frac{1}{0.757}(1.204)(1.628)} = 895 \text{ MPa} \tag{45}$$

Contact strength and allowable:

$$S_c = 2.22\,HB + 200 = 2.22(450) + 200 = 1199 \text{ MPa} \tag{46}$$

$$\sigma_{c,\text{allow}} = \frac{S_c \, Z_N C_H}{K_T K_R} = \frac{1199(1.600)(1)}{(1)(1)} = 1918 \text{ MPa} \tag{47}$$

$$n_c = \frac{\sigma_{c,\text{allow}}}{\sigma_c} = \frac{1918}{895} = 2.14 \tag{48}$$

Gear 1 has AGMA bending FOS $n_b = 2.72$ and contact FOS $n_c = 2.14$, both above the minimum, with infinite life.

one is wrong

## A.3 Bearing Selection

Both bearings on Shaft A are deep-groove ball bearings carrying pure radial load (spur gears, $F_a = 0$). Because the gear is close to the input bearing, the near bearing carries a much higher load than the far bearing, so the two are sized independently: a larger SKF 6409 near the input and a smaller SKF 6209 at the far end. Both share a ∅45 bore, so the shaft bearing seats are the same diameter.

**Reaction Forces (from shaft analysis):**

$$F_{r1} = 4704 \text{ N (near)} \qquad F_{r2} = 1731 \text{ N (far)} \qquad F_a = 0 \tag{49}$$

**Near Bearing — SKF 6409:**

$$P = F_{r1} = 4704 \text{ N} \qquad L_{10} = \frac{60 n L_h}{10^6} = 2271 \text{ million rev} \tag{50}$$

$$C_{req} = P\,(L_{10})^{1/3} = 4704(2271)^{1/3} = 61.8 \text{ kN} \tag{51}$$

$$\text{SKF 6409: ID 45, OD 120, } W \text{ 29 mm}, \quad C = 76.1 \text{ kN}, \; C_0 = 45 \text{ kN} \tag{52}$$

$$L_h = \left(\frac{C}{P}\right)^3 \frac{10^6}{60n} = \left(\frac{76100}{4704}\right)^3 \frac{10^6}{60(2600)} = 27{,}100 \text{ hr} \tag{53}$$

**Far Bearing — SKF 6209:**

$$P = F_{r2} = 1731 \text{ N} \qquad C_{req} = 1731(2271)^{1/3} = 22.8 \text{ kN} \tag{54}$$

$$\text{SKF 6209: ID 45, OD 85, } W \text{ 19 mm}, \quad C = 33.2 \text{ kN}, \; C_0 = 21.6 \text{ kN} \tag{55}$$

$$L_h = \left(\frac{33200}{1731}\right)^3 \frac{10^6}{60(2600)} = 45{,}200 \text{ hr} \tag{56}$$

**Result:** Both bearings exceed the required 14 560 hr with no changes needed (27 100 hr near, 45 200 hr far).

Blah



# Appendix B — Shaft System B

*Crystal Cooper*

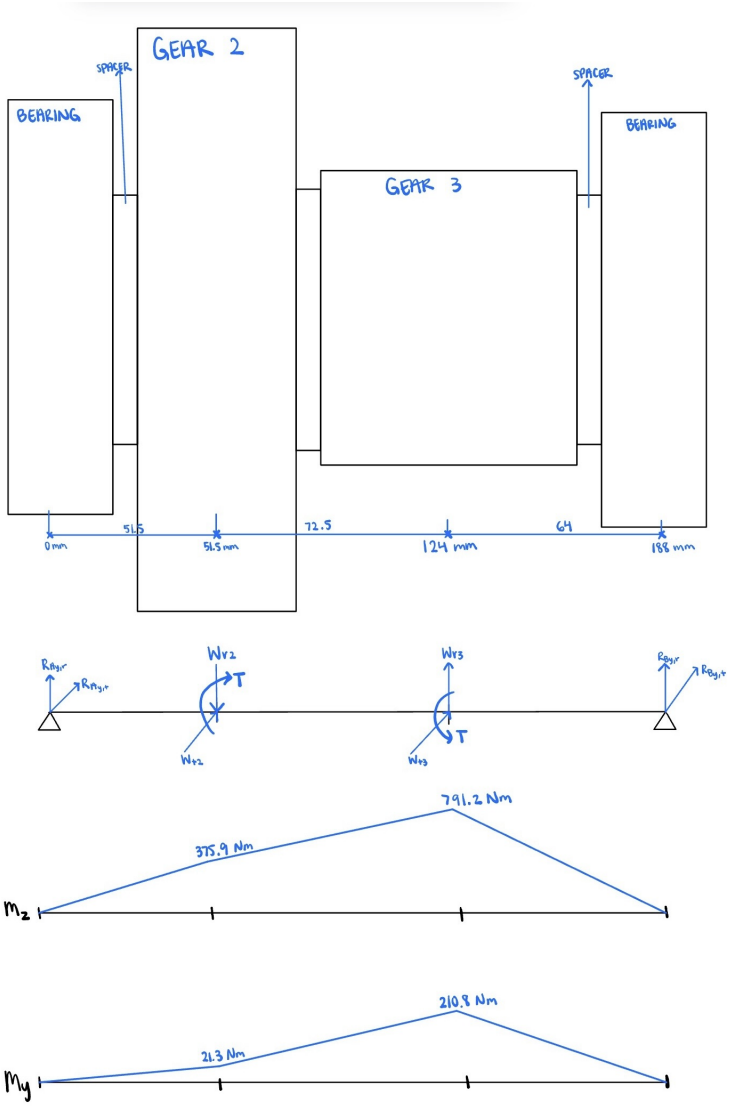



Figure 2: Shaft B free body diagram, tangential plane bending moment diagram ($M_z$), and radial plane bending moment diagram ($M_y$). Torque input (CW) at Gear 2 and output (CCW) at Gear 3. Critical section at Gear 3 keyway; resultant moment $M = \sqrt{791.2^2 + 210.8^2} = 818.8$ N·m.

## B.1 Shaft Analysis

Shaft B is the middle shaft, transmitting power between Shaft A and Shaft C at 866.7 rpm. It carries Gear 2 (51 teeth, first pair) and Gear 3 (19 teeth, second pair). Gear 2 sits 51.5 mm from Bearing 1 center; Gear 3 sits 124 mm from Bearing 1 center (bearing center-to-center span 188 mm; bearing inner faces 155 mm apart to match the gearbox housing). A shaft shoulder is placed between the two gears. The material is AISI 1045 cold-drawn steel. Because Shaft A sits above and Shaft C below Shaft B, the two radial gear forces act in opposite vertical directions. Shaft B rotates clockwise, driven by Gear 1 on Shaft A (counter-clockwise).

**Given:**

$$N_2 = 51 \qquad N_3 = 19 \qquad n_B = 866.7 \text{ rpm} \qquad H = 70 \text{ kW} \qquad m = 5 \text{ mm} \qquad \phi = 20^\circ \tag{57}$$

$$S_{ut} = 550 \text{ MPa} \qquad S_y = 310 \text{ MPa} \qquad n_s \geq 1.3 \tag{58}$$

**Torque:**

$$T = \frac{60H}{2\pi n_B} = \frac{60(70{,}000)}{2\pi(866.7)} = 771 \text{ N} \cdot \text{m} \tag{59}$$

**Pitch Diameters and Gear Forces:**

$$d_2 = N_2\, m = 51(5) = 255 \text{ mm} \qquad d_3 = N_3\, m = 19(5) = 95 \text{ mm} \tag{60}$$

$$W_{t2} = \frac{2T_A}{d_2} = \frac{2(257)}{0.255} = 6{,}047.1 \text{ N} \qquad W_{r2} = W_{t2} \tan\phi = 2{,}201 \text{ N (downward)} \tag{61}$$

$$W_{t3} = \frac{2T_B}{d_3} = \frac{2(771)}{0.095} = 16{,}231.6 \text{ N} \qquad W_{r3} = W_{t3} \tan\phi = 5{,}907.8 \text{ N (upward)} \tag{62}$$

**Shaft Geometry:**

$$L = d_{ii} + b_w = 155 + 33 = 188 \text{ mm} \tag{63}$$

$$a_2 = \frac{b_w}{2} + 10 + 25 = 16.5 + 10 + 25 = 51.5 \text{ mm} \qquad a_3 = L - \left(\frac{b_w}{2} + 10 + 37.5\right) = 188 - 64 = 124 \text{ mm} \tag{64}$$

**Bearing Reaction Forces — Tangential Plane:**

$$\sum M_{A,t} = 0: \quad R_{By,t}(188) = W_{t2}(51.5) + W_{t3}(124) \tag{65}$$

$$R_{By,t} = \frac{6{,}047.1(51.5) + 16{,}231.6(124)}{188} = 12{,}362.5 \text{ N} \tag{66}$$

$$R_{Ay,t} = W_{t2} + W_{t3} - R_{By,t} = 6{,}047.1 + 16{,}231.6 - 12{,}362.5 = 9{,}916.2 \text{ N} \tag{67}$$

**Bearing Reaction Forces — Radial Plane:**

Note: $W_{r2}$ acts downward (Gear 1 on Shaft A pushes Gear 2 down); $W_{r3}$ acts upward (Gear 4 on Shaft C pushes Gear 3 up).

$$\sum M_{A,r} = 0: \quad R_{By,r}(188) = W_{r2}(51.5) + W_{r3}(124) \tag{68}$$

$$R_{By,r} = \frac{2{,}201(51.5) + 5{,}907.8(124)}{188} = 3{,}293.7 \text{ N} \tag{69}$$

$$R_{Ay,r} = W_{r2} + W_{r3} - R_{By,r} = 2{,}201 - 5{,}907.8 + 3{,}293.7 = -413.1 \text{ N} \tag{70}$$

The negative sign for $R_{Ay,r}$ indicates Bearing 1 reacts downward in the radial plane due to the dominant upward $W_{r3}$.

**Resultant Radial Loads:**

$$F_{rA} = \sqrt{R_{Ay,t}^2 + R_{Ay,r}^2} = \sqrt{9{,}916.2^2 + 413.1^2} = 9{,}924.8 \text{ N} \tag{71}$$

$$F_{rB} = \sqrt{R_{By,t}^2 + R_{By,r}^2} = \sqrt{12{,}362.5^2 + 3{,}293.7^2} = 12{,}793.7 \text{ N} \tag{72}$$

Bearing B governs: $F_{rB} = 12\,793.7\,\text{N}$.

**Bending Moments at Critical Section (Gear 3):**

$$M_{t,G3} = R_{Ay,t}\, a_3 - W_{t2}(a_3 - a_2) = 9{,}916.2(0.124) - 6{,}047.1(0.0725) = 791.2 \text{ N} \cdot \text{m} \tag{73}$$

$$M_{r,G3} = R_{Ay,r}\, a_3 - W_{r2}(a_3 - a_2) = -413.1(0.124) - 2{,}201(0.0725) = -210.8 \text{ N} \cdot \text{m} \tag{74}$$

$$M_{res} = \sqrt{M_{t,G3}^2 + M_{r,G3}^2} = \sqrt{791.2^2 + 210.8^2} = 818.8 \text{ N} \cdot \text{m} \tag{75}$$

**Endurance Limit (Marin factors, machined surface):**

$$S_e' = 0.5\, S_{ut} = 275 \text{ MPa} \qquad k_a = 1.58\, S_{ut}^{-0.085} = 0.924 \tag{76}$$

$$k_b = 1.189\, d^{-0.112} = 1.189(65)^{-0.112} = 0.745 \qquad k_r = 0.82 \ (99\%) \qquad k_t = k_m = 1.0 \tag{77}$$

$$S_e = k_a k_b k_r k_t k_m S_e' = 0.924(0.745)(0.82)(1)(1)(275) = 155.2 \text{ MPa} \tag{78}$$

**Stress Concentration (keyway, AISI 1045):**

$$K_f = 1.61 \quad \text{(bending)} \qquad K_{fs} = 1.3 \quad \text{(torsion)} \tag{79}$$

From Spotts & Shoup (1998) for medium-carbon steel (SAE 1045, $S_{ut} \approx 550\,\text{MPa}$), per the project guidelines.

**Critical Section — Gear 3 Keyway (⌀65):**

$$\sigma_{xa} = \frac{32\, M_{res}}{\pi\, d_s^3} = \frac{32(818.8)}{\pi(0.065)^3} = 30.4 \text{ MPa} \qquad \sigma_{xm} = 0 \quad \text{(spur gear, no axial load)} \tag{80}$$

$$\tau_{xym} = \frac{16T}{\pi\, d_s^3} = \frac{16(771)}{\pi(0.065)^3} = 14.3 \text{ MPa} \tag{81}$$

$$\sigma_{VM,a} = \sqrt{(K_f\, \sigma_{xa})^2} = 1.61(30.4) = 48.9 \text{ MPa} \tag{82}$$

$$\sigma_{VM,m} = \sqrt{3\,(K_{fs}\, \tau_{xym})^2} = \sqrt{3}\,(1.3)(14.3) = 32.2 \text{ MPa} \tag{83}$$

**Goodman Fatigue Factor of Safety:**

$$\frac{1}{n_f} = \frac{\sigma_{VM,a}}{S_e} + \frac{\sigma_{VM,m}}{S_{ut}} = \frac{48.9}{155.2} + \frac{32.2}{550} = 0.3150 + 0.0585 = 0.3735 \;\Rightarrow\; n_f = 2.68 \tag{84}$$

**Yield Factor of Safety:**

$$\sigma_{VM,\max} = \sqrt{(K_f\,\sigma_{xa})^2 + 3(K_{fs}\,\tau_{xym})^2} = \sqrt{48.9^2 + 3(18.6)^2} = 58.5 \text{ MPa} \tag{85}$$

$$n_y = \frac{S_y}{\sigma_{VM,\max}} = \frac{310}{58.5} = 5.30 \tag{86}$$

**Governing Result:**

| | Goodman $n_f$ | Yield $n_y$ |
|---|---|---|
| Gear 3 keyway (⌀65) | 2.68 | 5.30 |

(87)

Both factors of safety exceed the minimum $n_s = 1.3$. The shaft has infinite life.

## B.2 Gear Analysis (Gears 2 and 3, AGMA)

Gears 2 and 3 are spur gears on Shaft B, analyzed by the AGMA method for bending and contact (pitting) with 99% reliability. Gear 2 (51 teeth, $b_w = 50\,\text{mm}$) is AISI 1045 CD at ∼200 HB. Gear 3 (19 teeth, $b_w = 75\,\text{mm}$) is AISI 1045 induction-hardened to ∼450 HB — required because its small tooth count and high load produce elevated bending stress.

**Life and Stress-Cycle Factors (both gears, $n_B = 866.7\,\text{rpm}$):**

$$N = \frac{14{,}560(60)(866.7)}{10^6} = 757.5 \text{ million cycles} \tag{88}$$

$$Y_N = 1.6831\,N^{-.0323} = 1.6831(757.5{\times}10^6)^{-.0323} = 0.87 \qquad Z_N = 2.466\,N^{-.056} = 2.466(757.5{\times}10^6)^{-.056} = 0.78 \tag{89}$$

**Common Factors:**

$$K_T = 1.0 \quad (T < 125^\circ\text{C}) \qquad K_R = 1.0 \quad (99\%\text{ reliability}) \qquad K_a = K_s = K_i = K_b = 1.0 \qquad K_e = 191MPa \tag{90}$$

**Gear 2** ($N_2 = 51$, $d_{p2} = 255\,\text{mm}$, $b_w = 50\,\text{mm}$, HB = 200, $W_t = 6047.1\,\text{N}$)

$$S_b = 0.533(200) + 88.3 = 194.9 \text{ MPa} \qquad S_c = 2.22(200) + 200 = 644.0 \text{ MPa} \tag{91}$$

$$\sigma_{all,b} = \frac{S_b\,Y_N}{K_T K_R} = \frac{194.9(0.870)}{1} = 169.6 \text{ MPa} \qquad \sigma_{all,c} = \frac{S_c\,Z_N\,C_H}{K_T K_R} = \frac{644.0(0.780)(1.01)}{1} = 507.2 \text{ MPa} \tag{92}$$

Pitch line velocity and dynamic factor ($Q_v = 6$):

$$V_2 = \pi\left(\frac{866.7}{60}\right)(0.255) = 11.6 \text{ m/s} \qquad B = 0.25(12{-}6)^{2/3} = 0.825 \qquad A = 50{+}56(1{-}0.825) = 59.8 \tag{93}$$

$$K_{v2} = \left(\frac{59.8 + \sqrt{200(11.6)}}{59.8}\right)^{0.825} = 1.59 \qquad K_{m2} = 1.2 \qquad Y_{J2} = 0.42 \qquad I_2 = 0.78 \tag{94}$$

Hardness ratio ($\text{HB}_{p1}/\text{HB}_{g2} = 450/200$):

$$C_H = 1 + 0.00698\left(\frac{51}{17} - 1\right) = 1.01 \tag{95}$$

Working stresses:

$$\sigma_{b2} = \frac{W_{t2}\, K_a K_s K_{m2} K_{v2}}{b_w\, m\, Y_{J2}} = \frac{6{,}047.1(1)(1)(1.2)(1.59)}{(0.050)(0.005)(0.42)} = 109.9 \text{ MPa} \tag{96}$$

$$\sigma_{c2} = K_e\sqrt{\frac{W_{t2}}{b_w\, d_{p2}} \cdot \frac{1}{I_2} \cdot K_a K_s K_{m2} K_{v2}} = 191\sqrt{\frac{6{,}047.1}{(0.050)(0.255)} \cdot \frac{1}{0.78} \cdot (1.2)(1.59)} = 205.7 \text{ MPa} \tag{97}$$

Factors of safety:

$$n_{s,b2} = \frac{\sigma_{all,b}}{\sigma_{b2}} = \frac{169.6}{109.9} = 1.54 \quad \checkmark \qquad n_{s,c2} = \frac{\sigma_{all,c}}{\sigma_{c2}} = \frac{507.2}{205.7} = 2.47 \quad \checkmark \tag{98}$$

**Gear 3** ($N_3 = 19$, $d_{p3} = 95\,\text{mm}$, $b_w = 75\,\text{mm}$, HB = 450 induction-hardened, $W_t = 16\,231.6\,\text{N}$)

$$S_b = 0.533(450) + 88.3 = 328.2 \text{ MPa} \qquad S_c = 2.22(450) + 200 = 1{,}199.0 \text{ MPa} \tag{99}$$

$$\sigma_{all,b} = \frac{328.2(0.870)}{1} = 285.5 \text{ MPa} \qquad \sigma_{all,c} = \frac{1{,}199.0(0.780)(1.01)}{1} = 944.7 \text{ MPa} \tag{100}$$

Pitch line velocity and dynamic factor ($Q_v = 6$):

$$V_3 = \pi\left(\frac{866.7}{60}\right)(0.095) = 4.32 \text{ m/s} \qquad K_{v3} = \left(\frac{59.8 + \sqrt{200(4.32)}}{59.8}\right)^{0.825} = 1.37 \tag{101}$$

$$K_{m3} = 1.1 \qquad Y_{J3} = 0.31 \qquad I_3 = 0.75 \qquad C_H = 1 + 0.00698\left(\frac{55}{19} - 1\right) = 1.01 \tag{102}$$

Working stresses:

$$\sigma_{b3} = \frac{W_{t3}\, K_a K_s K_{m3} K_{v3}}{b_w\, m\, Y_{J3}} = \frac{16{,}231.6(1)(1)(1.1)(1.37)}{(0.075)(0.005)(0.31)} = 210.6 \text{ MPa} \tag{103}$$

$$\sigma_{c3} = K_e\sqrt{\frac{W_{t3}}{b_w\, d_{p3}} \cdot \frac{1}{I_3} \cdot K_a K_s K_{m3} K_{v3}} = 191\sqrt{\frac{16{,}231.6}{(0.075)(0.095)} \cdot \frac{1}{0.75} \cdot (1.1)(1.37)} = 408.8 \text{ MPa} \tag{104}$$

Factors of safety:

$$n_{s,b3} = \frac{\sigma_{all,b}}{\sigma_{b3}} = \frac{285.5}{210.6} = 1.36 \quad \checkmark \qquad n_{s,c3} = \frac{\sigma_{all,c}}{\sigma_{c3}} = \frac{944.7}{408.8} = 2.31 \quad \checkmark \tag{105}$$

All bending and contact factors of safety exceed the minimum $n_s \geq 1.27$, confirming both gears have adequate strength with infinite life.



## B.3 Bearing Selection

Both bearings on Shaft B are single row deep-groove ball bearings (SKF 6313) carrying pure radial load (spur gears, $W_a = 0$). Bearing B carries the higher resultant load and governs the selection. The same designation is used for both positions to maintain consistent housing bore diameters on each side wall. Because the required dynamic load rating exceeds what a ball bearing at 50 mm bore can provide, the shaft bearing seats were stepped up to 65 mm.

**Reaction Forces (from shaft analysis):**

$$F_{rA} = 9{,}924.8 \text{ N} \qquad F_{rB} = 12{,}793.7 \text{ N (governs)} \qquad F_a = 0 \tag{106}$$

**Equivalent Dynamic Load:**

$$P = X\,F_{rB} + Y\,F_a = 1.0(12{,}793.7) + 0 = 12{,}793.7 \text{ N} \tag{107}$$

**Required Life and Dynamic Load Rating:**

One bearing change is planned at the 7280 hr midpoint (half of the 14 560 hr duty life), so each bearing interval is 7280 hr.

$$L_{10} = \frac{60\,n_B\,L_h}{10^6} = \frac{60(866.7)(7{,}280)}{10^6} = 378.6 \text{ million rev} \tag{108}$$

$$C_{req} = P\,(L_{10})^{1/3} = 12{,}793.7\,(378.6)^{1/3} = 12{,}793.7(7.238) = 92.6 \text{ kN} \tag{109}$$

**Selected Bearing — SKF 6313 (both positions):**

$$\text{ID} = 65 \text{ mm}, \quad \text{OD} = 140 \text{ mm}, \quad W = 33 \text{ mm}, \quad C = 97.5 \text{ kN}, \quad C_0 = 60 \text{ kN} \tag{110}$$

$$C = 97.5 \text{ kN} \;\geq\; C_{req} = 92.6 \text{ kN} \quad \checkmark \tag{111}$$

**Actual Bearing Life per Interval:**

$$L_{10,act} = \left(\frac{C}{P}\right)^3 = \left(\frac{97{,}500}{12{,}793.7}\right)^3 = 442.6 \text{ million rev} \tag{112}$$

$$L_{h,act} = \frac{442.6 \times 10^6}{60(866.7)} = 8{,}511 \text{ hr} \qquad L_{y,act} = \frac{8{,}511}{8 \times 365} = 2.91 \text{ years per interval} \tag{113}$$

**Result:** With one planned replacement at approximately year 3, the total bearing service covers the full 14 560 hr (5-year) duty life. Each SKF 6313 bearing provides 8511 hr of life under the governing load, confirming the replacement schedule is sufficient.



# Appendix C — Shaft System C

*Sujee Rubio*

## C.1 Shaft Analysis

Shaft C is the output shaft, delivering power through a coupling at 299.4 rpm. It carries Gear 4, the 55-tooth gear of the second gear pair. The gear center sits 129 mm from Bearing 1 and 59 mm from Bearing 2 (center-to-center span 188 mm). Material is AISI 1045 cold-drawn steel. The shaft diameter at the critical section (gear seat) is 70 mm; bearing seats step down to 65 mm with a locating shoulder.

**Given:**

$$N_4 = 55 \qquad n_C = 299.4 \text{ rpm} \qquad H = 70 \text{ kW} \qquad m = 5 \text{ mm} \qquad \phi = 20^\circ \tag{114}$$

$$S_{ut} = 550 \text{ MPa} \qquad S_y = 310 \text{ MPa} \qquad n_s \geq 1.3 \tag{115}$$

**Torque:**

$$T = \frac{60H}{2\pi n_C} = \frac{60(70{,}000)}{2\pi(299.4)} = 2233 \text{ N} \cdot \text{m} \tag{116}$$

**Pitch Diameter and Gear Forces:**

$$d_4 = N_4\, m = 55(5) = 275 \text{ mm} \tag{117}$$

$$W_t = \frac{2T}{d_4} = \frac{2(2233)}{0.275} = 16{,}231.6 \text{ N} \qquad W_r = W_t \tan\phi = 5{,}907.8 \text{ N} \qquad W_a = 0 \tag{118}$$

**Bearing Reaction Forces:** With $a = 129$ mm, $b = 59$ mm, $L = 188$ mm:

$$R_{1t} = W_t \frac{b}{L} = 16{,}231.6 \left(\frac{59}{188}\right) = 5{,}094.0 \text{ N} \qquad R_{2t} = W_t \frac{a}{L} = 11{,}137.6 \text{ N} \tag{119}$$

$$R_{1r} = W_r \frac{b}{L} = 5{,}907.8 \left(\frac{59}{188}\right) = 1{,}854.0 \text{ N} \qquad R_{2r} = W_r \frac{a}{L} = 4{,}053.8 \text{ N} \tag{120}$$

$$F_{r1} = \sqrt{R_{1t}^2 + R_{1r}^2} = \sqrt{5094^2 + 1854^2} = 5{,}420.9 \text{ N} \qquad F_{r2} = \sqrt{R_{2t}^2 + R_{2r}^2} = 11{,}852.4 \text{ N} \tag{121}$$

Bearing 2 governs: $F_{r2} = 11\,852.4$ N.

**Bending Moments at Critical Section (Gear 4):**

$$M_t = R_{1t}\, a = 5094.0(0.129) = 657.1 \text{ N} \cdot \text{m} \qquad M_r = R_{1r}\, a = 1854.0(0.129) = 239.2 \text{ N} \cdot \text{m} \tag{122}$$

$$M = \sqrt{M_t^2 + M_r^2} = \sqrt{657.1^2 + 239.2^2} = 699.3 \text{ N} \cdot \text{m} \tag{123}$$

**Endurance Limit (Marin factors, machined surface):**

$$S_e' = 0.5\, S_{ut} = 275 \text{ MPa} \qquad k_a = 1.58\, S_{ut}^{-0.085} = 0.924 \tag{124}$$

$$k_b = 1.189\, d^{-0.112} = 1.189(70)^{-0.112} = 0.739 \qquad k_r = 0.82 \ (99\%) \qquad k_t = k_m = 1.0 \tag{125}$$

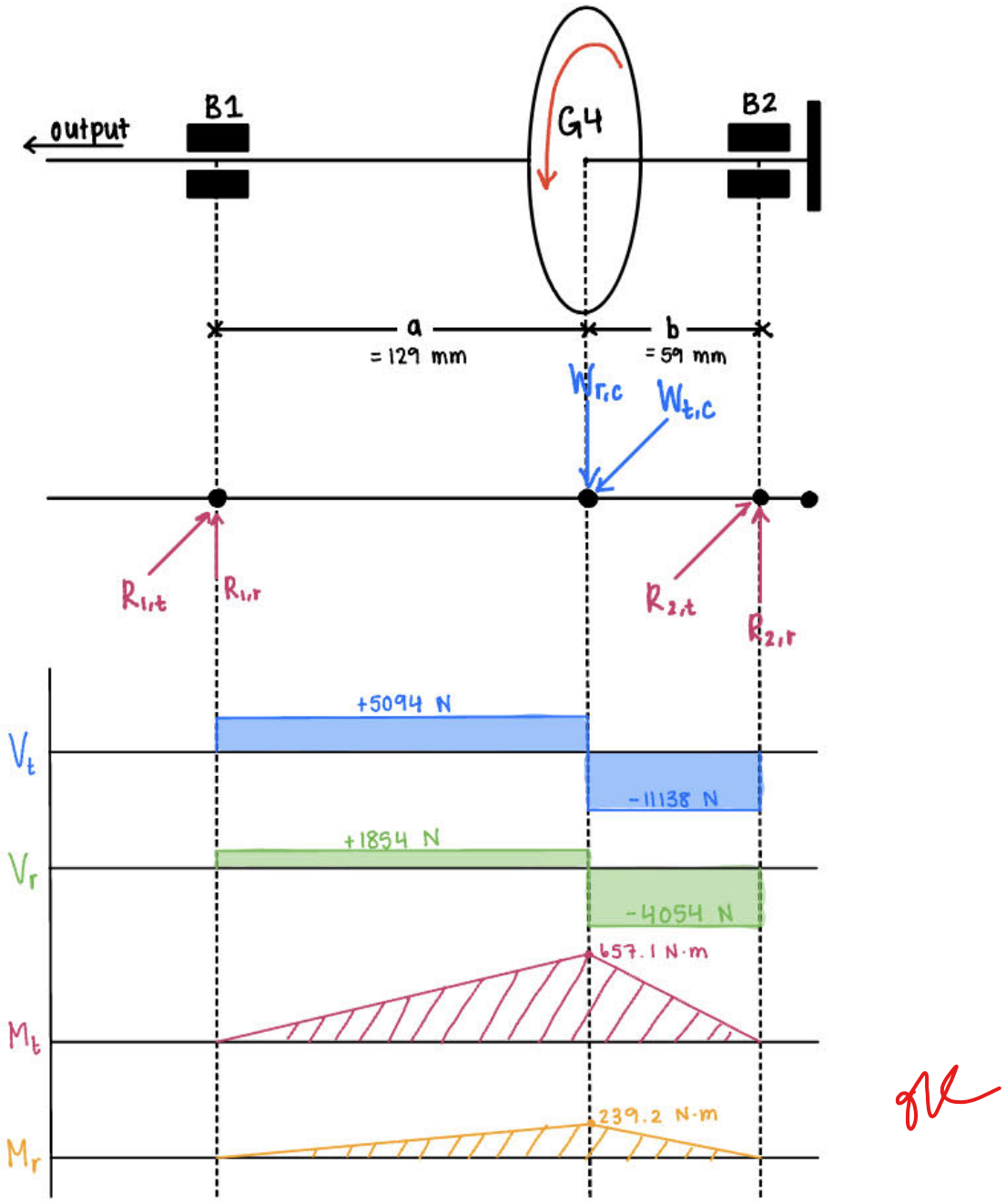



Figure 3: Shaft C FBD, shear force, and bending moment diagrams.

$$S_e = k_a k_b k_r k_t k_m S'_e = 0.924(0.739)(0.82)(1)(1)(275) = 154.0 \text{ MPa} \tag{126}$$

**Stress Concentration (keyway, AISI 1045):**

$$K_f = 2.0 \quad \text{(bending)} \qquad K_{fs} = 1.6 \quad \text{(torsion)} \tag{127}$$

From Spotts & Shoup (1998) for medium-carbon steel (SAE 1045, $S_{ut} \approx 550\,\text{MPa}$). Conservative values used to account for both the gear keyway and any shoulder effects.

**Two critical cross sections are checked: the gear keyway (⌀70) and the coupling keyway (⌀60). The gear section carries the maximum bending moment; the coupling carries torsion only but at a smaller diameter, producing higher torsional stress.**

**Critical Section 1 — Gear Keyway (⌀70 mm):**

$$\sigma_{xa} = \frac{32M}{\pi d^3} = \frac{32(699.3)}{\pi(0.070)^3} = 20.77 \text{ MPa} \qquad \sigma_{xm} = 0 \quad \text{(spur gear, no axial load)} \tag{128}$$

$$\tau_{xym} = \frac{16T}{\pi d^3} = \frac{16(2233)}{\pi(0.070)^3} = 33.16 \text{ MPa} \tag{129}$$

$$\sigma_{VM,a} = \sqrt{(K_f\,\sigma_{xa})^2} = 2.0(20.77) = 41.53 \text{ MPa} \tag{130}$$

$$\sigma_{VM,m} = \sqrt{\sigma_{xm}^2 + 3(K_{fs}\,\tau_{xym})^2} = \sqrt{0 + 3(1.6 \times 33.16)^2} = 91.89 \text{ MPa} \tag{131}$$

$$\frac{1}{n_f} = \frac{\sigma_{VM,a}}{S_e} + \frac{\sigma_{VM,m}}{S_{ut}} = \frac{41.53}{154.0} + \frac{91.89}{550} = 0.2697 + 0.1671 = 0.4368 \;\Rightarrow\; n_f = 2.29 \tag{132}$$

Yield check (von Mises at peak load):

$$\sigma_{VM,\max} = \sqrt{(K_f\,\sigma_{xa} + \sigma_{xm})^2 + 3(K_{fs}\,\tau_{xym})^2} = \sqrt{(41.53)^2 + 3(53.06)^2} = 100.84 \text{ MPa} \tag{133}$$

$$n_y = \frac{S_y}{\sigma_{VM,\max}} = \frac{310}{100.84} = 3.07 \tag{134}$$

**Critical Section 2 — Coupling Keyway (⌀60 mm):** At the coupling end there is no bending ($\sigma_{xa} = 0$); only torsion acts. The smaller diameter raises the torsional stress relative to the gear section.

$$\tau_{xym} = \frac{16T}{\pi d^3} = \frac{16(2233)}{\pi(0.060)^3} = 52.57 \text{ MPa} \tag{135}$$

$$\sigma_{VM,a} = 0 \qquad \sigma_{VM,m} = \sqrt{3}\,K_{fs}\,\tau_{xym} = \sqrt{3}(1.6)(52.57) = 145.7 \text{ MPa} \tag{136}$$

$$\frac{1}{n_f} = \frac{0}{S_e} + \frac{145.7}{550} \;\Rightarrow\; n_f = 3.77 \tag{137}$$

Yield check:

$$\sigma_{VM,\max} = \sqrt{3}\,K_{fs}\,\tau_{xym} = 145.7 \text{ MPa} \tag{138}$$

$$n_y = \frac{S_y}{\sigma_{VM,\max}} = \frac{310}{145.7} = 2.13 \tag{139}$$

**Governing Result:**

| | Goodman $n_f$ | Yield $n_y$ |
|---|---|---|
| Gear keyway (⌀70) | 2.29 | 3.07 |
| Coupling keyway (⌀60) | 3.77 | 2.13 |

(140)

The gear keyway governs on Goodman ($n_f = 2.29$) and the coupling keyway governs on yield ($n_y = 2.13$). Both governing values exceed the minimum $n_s = 1.3$. The shaft has infinite life.

**Keyway Design:** Parallel (rectangular) keys are selected per DIN 6885. Two keyways are required: one at the gear seat and one at the output coupling.

| | Shaft $d$ (mm) | $W$ (mm) | $H$ (mm) | Depth $H/2$ (mm) |
|---|---|---|---|---|
| Gear seat | 70 | 20 | 12 | 6.0 |
| Coupling seat | 60 | 18 | 11 | 5.5 |

(141)

Key lengths: gear keyway $L_k = 75\,\text{mm}$ (equal to face width); coupling keyway $L_k = 50\,\text{mm}$ (matched to hub length). Both keyways are end-mill cut with rounded ends and a root fillet $r \approx 0.3\,\text{mm}$. The stress analysis above used $K_f = 2.0$ and $K_{fs} = 1.6$, which conservatively account for keyway stress concentration; no shaft resizing is required.

## C.2 Gear Analysis (Gear 4, AGMA)

Gear 4 is the 55-tooth spur gear of the second pair (AISI 1045 CD, $\sim$200 HB). Module 5, pressure angle 20°, face width 75 mm. Analyzed by the AGMA method for both bending and contact (pitting) with 99% reliability.

**Life and Stress-Cycle Factors:**

$$N_{C4} = \frac{14{,}560(60)(299.4)}{10^6} = 261.6 \text{ million cycles} \tag{142}$$

$$Y_{N4} = 1.6831\, N^{-0.0323} = 1.6831(261.6 \times 10^6)^{-0.0323} = 1.406 \tag{143}$$

$$Z_{N4} = 2.466\, N^{-0.056} = 2.466(261.6 \times 10^6)^{-0.056} = 1.806 \tag{144}$$

**Pitch Line Velocity and Dynamic Factor ($Q_v = 6$):**

$$V = \pi\, n_C\, d_{p4} = \pi \left(\frac{299.4}{60}\right)(0.275) = 4.31 \text{ m/s} \tag{145}$$

$$K_v = 1.39 \quad \text{(from AGMA chart, } Q_v = 6) \tag{146}$$

**Stress and Load Factors:**

$$K_a = 1.0 \quad K_s = 1.0 \quad K_m = 1.1 \quad K_i = 1.0 \quad K_b = 1.0 \tag{147}$$

Hardness ratio: $C_H = 1.0$ (same material pinion and gear). Temperature factor: $K_T = 1.0$ ($T < 125$°C). Reliability factor: $K_R = 1.0$ (99%).

**Geometry Factors (AGMA chart, $N_4 = 55$, $N_3 = 19$):**

$$Y_{J4} = 0.43 \qquad I = \frac{\pi \cos\phi \, \sin\phi}{2} \cdot \frac{1}{1 + d_{p3}/d_{p4}} = \frac{\pi \cos 20^\circ \sin 20^\circ}{2} \cdot \frac{1}{1 + 95/275} = 0.375 \qquad K_e = 191 \text{ MPa} \tag{148}$$

**Allowable Stresses:**

$$S_b = 0.533\,\text{HB}{+}88.3 = 0.533(200){+}88.3 = 194.9 \text{ MPa} \qquad S_c = 2.22\,\text{HB}{+}200 = 2.22(200){+}200 = 644.0 \text{ MPa} \tag{149}$$

$$\sigma_{all,b} = \frac{S_b\, Y_{N4}}{K_T\, K_R} = \frac{194.9(1.406)}{(1)(1)} = 137.0 \text{ MPa} \qquad \sigma_{all,c} = \frac{S_c\, Z_{N4}\, C_H}{K_T\, K_R} = \frac{644.0(1.806)(1)}{(1)(1)} = 1162.8 \text{ MPa} \tag{150}$$

**Working Bending Stress:**

$$\sigma_b = \frac{W_t}{b_w\, m\, Y_{J4}}\, K_a K_s K_m K_v = \frac{16{,}231.6}{(0.075)(0.005)(0.43)}(1.0)(1.0)(1.1)(1.39) = 99.7 \text{ MPa} \tag{151}$$

**Working Contact Stress:**

$$\sigma_c = K_e \sqrt{\frac{W_t}{b_w\, d_{p4}\, I}\, K_a K_s K_m K_v} = 191 \sqrt{\frac{16{,}231.6}{(0.075)(0.275)(0.375)}(1.0)(1.0)(1.1)(1.39)} = 342.1 \text{ MPa} \tag{152}$$

**Factors of Safety:**

$$n_{s,b} = \frac{\sigma_{all,b}}{\sigma_b} = \frac{137.0}{99.7} = 1.37 \qquad n_{s,c} = \frac{\sigma_{all,c}}{\sigma_c} = \frac{1162.8}{342.1} = 3.40 \tag{153}$$

Gear 4 has AGMA bending FOS $n_{s,b} = 1.37$ and contact FOS $n_{s,c} = 3.40$, both above the minimum of 1.27, with infinite life.

## C.3 Bearing Selection

Both bearings on Shaft C are deep-groove ball bearings carrying pure radial load (spur gear, $W_a = 0$). Because Bearing 2 carries a significantly higher load than Bearing 1, the same bearing designation (SKF 6313) is used for both positions to keep the housing bore diameters consistent on each side wall.

**Reaction Forces (from shaft analysis):**

$$F_{r1} = 5{,}420.9 \text{ N} \qquad F_{r2} = 11{,}852.4 \text{ N (governs)} \qquad F_a = 0 \tag{154}$$

**Equivalent Dynamic Load ($X$, $Y$ factors):** Shaft C carries a spur gear ($W_a = 0$), so there is no axial load. Check $P_a/C_0$ to determine $X$ and $Y$ from Table 6-2:

$$\frac{P_a}{C_0} = \frac{0}{60{,}000} = 0 < 0.025 \tag{155}$$

For $P_a/C_0 < 0.025$ and $P_a/P_r \le e$: $X = 1,\ Y = 0$. The equivalent dynamic load is therefore:

$$P = A_p(X\,P_r + Y\,P_a) = 1.0\big(1.0 \times 11{,}852.4 + 0\big) = 11{,}852.4 \text{ N} \tag{156}$$

**Required Life and Dynamic Load Rating:**

$$L_{10} = \frac{60\,n_C\,L_h}{10^6} = \frac{60(299)(14{,}560)}{10^6} = 261.6 \text{ million rev} \tag{157}$$

$$C_{req} = P\,(L_{10})^{1/3} = 11{,}852.4\,(261.6)^{1/3} = 75{,}800 \text{ N} = 75.8 \text{ kN} \tag{158}$$

**Selected Bearing — SKF 6313 (both positions):**

$$\text{ID} = 65 \text{ mm}, \quad \text{OD} = 140 \text{ mm}, \quad W = 33 \text{ mm}, \quad C = 97.5 \text{ kN}, \quad C_0 = 60 \text{ kN} \tag{159}$$

$$C = 97.5 \text{ kN} \;\ge\; C_{req} = 75.8 \text{ kN} \quad \checkmark \tag{160}$$

The bearing seat diameter is 65 mm, stepped down from the 70 mm gear seat via a 5 mm-wide locating shoulder.

**Actual Bearing Life — Bearing 1 ($F_{r1} = 5{,}420.9$ N):**

$$L_{10,1} = \left(\frac{C}{F_{r1}}\right)^3 = \left(\frac{97{,}500}{5{,}420.9}\right)^3 = 5{,}812 \text{ million rev} \tag{161}$$

$$L_{h,1} = \frac{5{,}812 \times 10^6}{60 \times 299} = 323{,}700 \text{ hr} \qquad L_{y,1} = \frac{323{,}700}{8 \times 365} = 110.8 \text{ years} \quad \Rightarrow \quad \text{No change needed} \tag{162}$$

**Actual Bearing Life — Bearing 2 ($F_{r2} = 11{,}852.4$ N):**

$$L_{10,2} = \left(\frac{C}{F_{r2}}\right)^3 = \left(\frac{97{,}500}{11{,}852.4}\right)^3 = 556.7 \text{ million rev} \tag{163}$$

$$L_{h,2} = \frac{556.7 \times 10^6}{60 \times 299} = 30{,}988 \text{ hr} \qquad L_{y,2} = \frac{30{,}988}{8 \times 365} = 10.6 \text{ years} \quad \Rightarrow \quad \text{No change needed} \tag{164}$$

**Result:** Both bearings exceed the required 5-year service life. Bearing 1 lasts 110.8 years and Bearing 2 lasts 10.6 years; no bearing changes are required within the design life.